AE6357 Lunar Reconnaissance Orbiter Report

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Contents

1 Introduction

Understanding and predicting the motion of satellites and celestial bodies with high accuracy is critical for mission planning, navigation, and scientific research in astrodynamics. Tools such as NASA's SPICE (Spacecraft Planet Instrument C-matrix Events) toolkit have become extremely useful for accessing and interpreting precise spacecraft and planetary ephemerides for both current mission purposes and independent research. In contrast, a custom-built numerical integration code is developed to simulate the same trajectory using initial state conditions and incorporating gravitational modeling through third-body perturbations and spherical harmonics. The goal of this study is to compare the outputs of these two approaches—SPICE's ephemeris-based method and the numerically integrated model—in terms of accuracy, computational performance, and practical implementation. By doing so, we aim to evaluate the advantages and limitations of using precomputed ephemeris data versus a fully dynamic propagation model for spacecraft trajectory analysis using information taught over the course of this semester.

2 LRO Mission Background

The Lunar Reconnaissance Orbiter (LRO) is a NASA-led robotic mission that was launched to gather data for future human and robotic exploration of the Moon. Developed by NASA's Goddard Space Flight Center and funded through NASA's Science Mission Directorate and Human Exploration and Operations Mission Directorate, LRO's primary objectives included identifying safe landing sites, locating potential lunar resources, mapping surface topography in high resolution, and studying the Moon's radiation environment to support future crewed missions. LRO was worked alongside the Lunar Crater Observation and Sensing Satellite (LCROSS), as part of the United States's Vision for Space Exploration program and on June 18, 2009, aboard an Atlas V 401 rocket from Cape Canaveral, Florida. It entered lunar orbit on June 23, 2009. As of 2025, LRO continues to operate in an extended mission phase, offering critical support to NASA's Artemis program and serving as a scout for future robotic and crewed landings. Its data is used not only by NASA but also by international space agencies, independent researchers, and educators worldwide [3].

3 True Data Processing

To obtain true data that will be used to compare the effectiveness of the propagated result, a publicly accessible NASA tool-kit was used to provide accurate satellite and planetary body positions and velocity within 1 meter which will be assumed as the truth state for both the initial and final times.

3.1 SPICE Toolkit

SPICE (an acronym for Spacecraft, Planet, Instrument, C-matrix, Events) is a system developed by NASA's Navigation and Ancillary Information Facility (NAIF) to support planetary mission planning, data analysis, and engineering tasks. It provides precise geometric and timing information about spacecraft, celestial bodies, instruments, and mission events through specialized data files called "kernels." These kernels contain ephemerides, orientation data, instrument specifications, and timing conversions, among other critical parameters. SPICE includes a software Toolkit with APIs that allow users to extract observation geometry and event timing, enabling independent researchers to analyze mission data, simulate scenarios, or correlate results across different missions. Extensively documented and designed for flexibility, SPICE supports a wide range of missions and is accessible to the global scientific community through NASA's archives [1]. While SPICE can be implemented in a number of different coding languages, this project utilizes the Matlab SPICE Toolkit, MICE.

3.2 Time and Reference Frames

The SPICE Toolkit supports conversions between multiple time systems, including spacecraft clock time (SCLK), Coordinated Universal Time (UTC), Ephemeris Time (ET, also known as Barycentric Dynamical Time or TDB), and others. These conversions rely on dedicated kernel files such as SCLK kernels for spacecraft time and LSK (leap seconds) kernels for accurate UTC-to-ET translations.

In terms of reference frames, SPICE allows users to work with a variety of inertial and body-fixed frames. These include standard celestial reference frames (e.g., J2000), planet-fixed frames (e.g., IAU

Earth), and instrument or spacecraft-specific frames defined in Frames (FK) kernels. SPICE ensures precise transformations between these frames using orientation data from C-matrix (CK) and planetary constants (PCK) kernels, enabling accurate positioning and pointing calculations across different observational and modeling contexts.

For the purposes of comparing information in a practical frame, both the SPICE and propagated code transforms all position and velocity vectors into the Moon Centered frame (defined as MOON ME in the SPICE functions). In addition, both approaches utilize the UTC time system. While data was provided for the operational lifetime of LRO up to 2024, data from the extended lifetime operation section of the LRO satellite was chosen as it could be expected that no major burn maneuvers were taking place in this time frame which could lead to significant error in final state truth comparisons if not accounted for. The most recent spk file for the LRO tracks its location over a 3 month window starting March 15th, 2024 0hr UTC and ending June 15th, 2024 0hr UTC.

4 Independent Code

For the independent propagation code, the numerical integrator function ode45 was used with inputs of the initial true position and velocity states as well as a time span set to cover the a 1 day window from March 15th to March 16th of the SPICE ephemeris data. Since ode45's time step can not be controlled or set directly and the time span is fairly large, it is possible to reduce computation time indirectly by reducing the tolerance. This may lead to slightly less accurate final results but greatly increases the computational speed. The tolerance ran when computing these results was 1e-13. The true state propagation from the SPICE files over the specified time frame is shown below and is what the independent code aims to achieve.

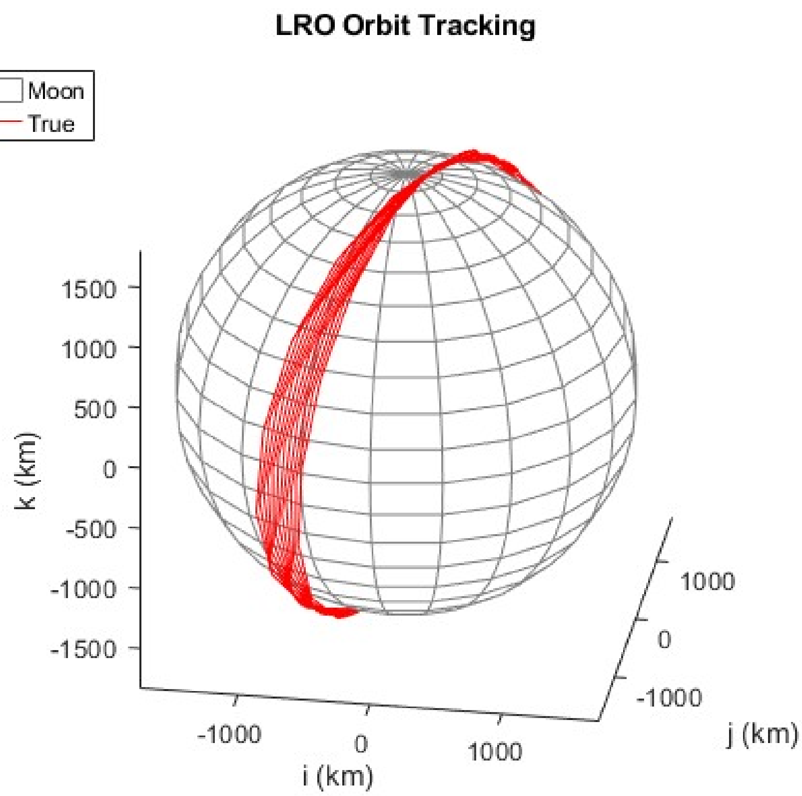

Figure 1: True LRO Orbit Propagation from SPICE toolkit

4.1 Force Modeling

In Newtonian physics, the motion of a satellite is determined by the total force acting upon it, expressed through a second-order differential equation involving the satellite's position, velocity, and mass. Throughout the course, the traditional dominant force was Earth's gravity but since this is a lunar mission, the main contributions from gravity are due to the Moon's gravitational field. Outside of the Moon's gravity contribution, effects on the satellites orbit due to third bodies, lunar spherical harmonics, and solar radiation pressure were also taken into account to get a more accurate final position and velocity prediction. All of the calculations outlined in the following sections were taken from Satellite Orbits by Montenbruck and Gill [5]. The LRO mission had an elliptic orbit with the following orbital parameters at the starting epoch which were used to model the effects of these additional forces on the satellite as a function of altitude from the moon. The figure below provides a good representation of what forces are of the most importance at a given altitude for a lunar orbit.

Table 1: Initial Orbital Elements

initial orbital elements	a	e	i	Ω	ω	f
value	1827.7 km	0.0078	83.9004 $^\circ$	106.134 $^\circ$	71.2 $^\circ$	-4.26 $^\circ$

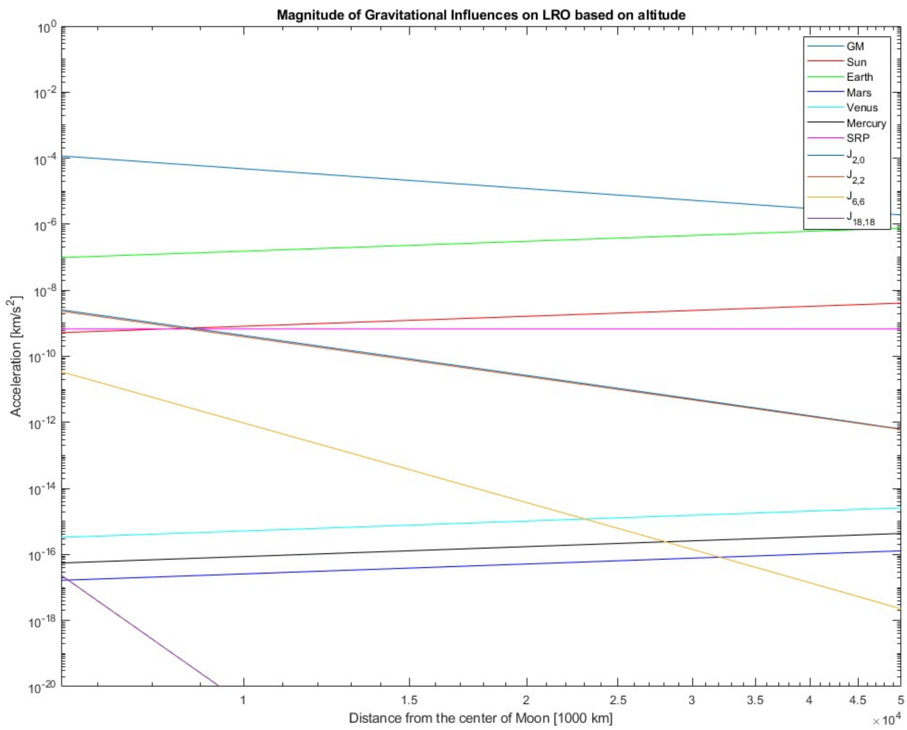

Figure 2: Force Modeling based on Altitude for a Lunar Orbit

Since the LRO maintained about a 50km altitude above the moon, one can see from the graph that the Earth and Moon forces are the most prominent and that the additional lunar gravitational terms rapidly drop off as will be discussed in the coming sections.

4.1.1 Third Body

The third body contributions considered were limited to the that of the planetary ephemeris data provided in the SPICE file used. For this particular LRO mission, third body effects due to the Earth, Sun, Mercury, Venus, and Mars were considered. The actual planetary position in the Moon ME frame as observed from the Moon was able to provide a good estimate of the satellites position with respect to the third body at each epoch. The following equation was used to calculate the acceleration of a satellite due to third body influence. where r and s are the geocentric coordinates of the satellite and of M, respectively.

$$\ddot{r} = GM\left(\frac{s-r}{|s-r|^3} - \frac{s}{|s|^3}\right) \tag{1}$$

4.1.2 Spherical Harmonics

Since the LRO would be orbiting the moon at a very low altitude and the moon has various craters with high and low density points, it was important to consider the spherical harmonics of the moon. This could be accessed from NASA's Planetary Data System website [2]. The lunar spherical harmonics data was collected during NASA's GRAIL (Gravity Recovery and Interior Laboratory) mission, which used two spacecraft equipped with the Lunar Gravity Ranging System (LGRS). These spacecraft worked in conjunction with Earth-based stations from the Deep Space Network (DSN) to precisely measure variations in the Moon's gravity. The data gathered enabled the creation of detailed, high-resolution spherical harmonic models that map the Moon's gravity field. Alongside these models, the data set includes covariance matrices and gravity maps for some versions. Additionally, it provides specialized SPICE kernel files, which offer improved spacecraft trajectory data developed by the GRAIL Science Data System (SDS), offering a more refined analysis than the standard navigation files from the mission.

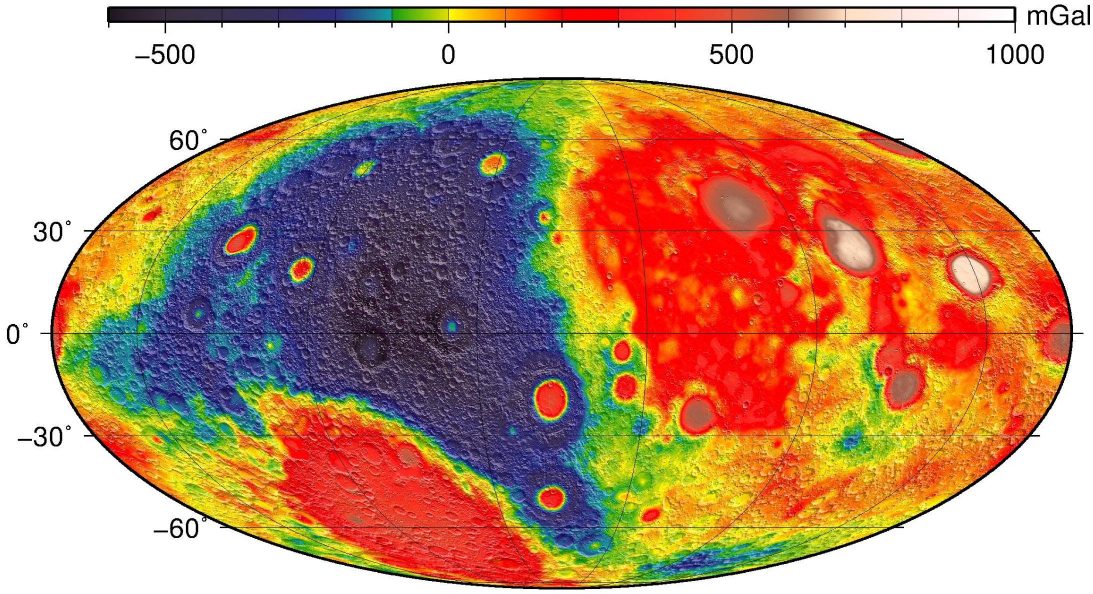

Figure 3: Force Modeling based on Altitude for a Lunar Orbit

The full 1800x1800 sha file was used in combination with the spherical harmonics equations for numerical integration outlined in the Satellite Orbits text to provide the most accurate simulation data. The lunar gravity field is shown in the below figure.

$$U = \frac{\mu_\oplus}{R_\oplus} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \left(C_{nm}V_{nm} + S_{nm}W_{nm}\right) \tag{2}$$

$$V_{nm} = \frac{2n-1}{n-m}\left(\frac{zR_\oplus}{r^2}\right)V_{(n-1),m} - \frac{n+m-1}{n-m}\left(\frac{R_\oplus^2}{r^2}\right)V_{(n-2),m} \tag{3}$$

$$W_{nm} = \frac{2n-1}{n-m}\left(\frac{zR_\oplus}{r^2}\right)W_{(n-1),m} - \frac{n+m-1}{n-m}\left(\frac{R_\oplus^2}{r^2}\right)W_{(n-2),m} \tag{4}$$

For $m = 0$ terms:

$$\ddot{x}_{nm} = \frac{\mu}{R_\oplus^2}\left(-C_{n0}V_{(n+1),1}\right) \tag{5}$$

$$\ddot{y}_{nm} = \frac{\mu}{R_\oplus^2}\left(-C_{n0}W_{(n+1),1}\right) \tag{6}$$

For $m > 0$ terms:

$$\ddot{x}_{nm} = \frac{\mu}{R_\oplus^2}\frac{1}{2}\left[\left(-C_{nm}V_{(n+1),(m+1)} - S_{nm}W_{(n+1),(m+1)}\right) + \frac{(n-m+2)!}{(n-m)!}\left(C_{nm}V_{(n+1),(m-1)} + S_{nm}W_{(n+1),(m-1)}\right)\right] \tag{7}$$

$$\ddot{y}_{nm} = \frac{\mu}{R_\oplus^2}\frac{1}{2}\left[\left(-C_{nm}W_{(n+1),(m+1)} + S_{nm}V_{(n+1),(m+1)}\right) + \frac{(n-m+2)!}{(n-m)!}\left(-C_{nm}W_{(n+1),(m-1)} + S_{nm}V_{(n+1),(m-1)}\right)\right] \tag{8}$$

The vertical component:

$$\ddot{z}_{nm} = \frac{\mu}{R_\oplus^2}(n-m+1)\left(-C_{nm}V_{(n+1),m} - S_{nm}W_{(n+1),m}\right) \tag{9}$$

$$\ddot{x} = \sum_{n,m}\ddot{x}_{nm}, \ddot{y} = \sum_{n,m}\ddot{y}_{nm}, \ddot{z} = \sum_{n,m}\ddot{z}_{nm} \tag{10}$$

4.1.3 Solar Radiation Pressure

Next, solar radiation pressure was taken into consideration along with the size of the LRO and the instruments used on board. The LRO weighed roughly 1000 kg and had a cross-sectional area of about 10.6 m^2 [4]. The LRO deisgn can be seen in the below figure.

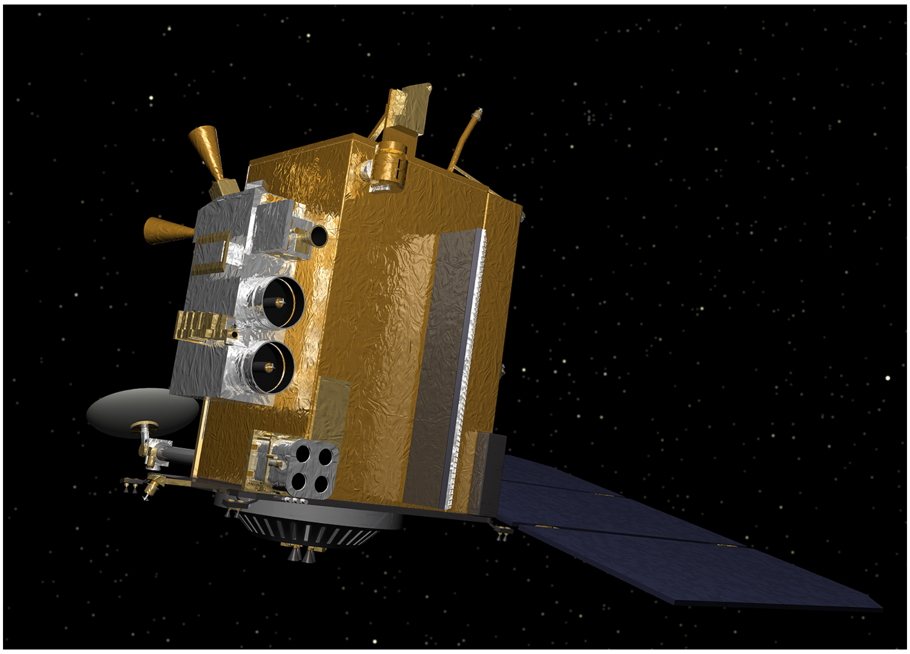

Figure 4: LRO Spacecraft Design

Since the LRO used both solar panels and a high gain antenna, the reflectivity (ε) was assumed to be 0.3 and thus the C_R for this spacecraft was assumed to be 1.30 as outline in the below table from the Satellite Orbits text.

Material	ε	$1-\varepsilon$	$C_R \approx 1+\varepsilon$
Solar panel	0.21	0.79	1.21
High-gain antenna	0.30	0.70	1.30
Aluminum coated mylar solar sail	0.88	0.12	1.88

Figure 5: Reflectivity, absorption and radiation pressure coefficient of selected satellite components [5]

For the calculation of acceleration due to solar radiation pressure, the following equation was used. It is important to note here that the solar radiation pressure for the satellite was assumed to be the same as that of the solar radiation pressure at the Earth ($P_{☼} \approx 4.56 \cdot 10^{16} Nm^{-2}$) since the distance from the moon to the sun and the Earth to the sun would be fairly similar by comparison.

$$\ddot{r} = -P_{☼} C_R \frac{A}{m} \frac{r_{☼}}{|r_{☼}|^3} AU^2 \tag{11}$$

5 Results

The true initial and final positions that independent code was trying to reach are provided in the table below.

Table 2: True Final and Initial Conditions of the LRO

TRUE	Date	LRO Position [km]	LRO Velocity [km/s]
initial	March 15th, 2024 00:00:00.000 UTC	[513.3, 1344.0, 1119.4]	[0.1533, 1.0209, -1.2765]
final	March 16th, 2024 00:00:00.000 UTC	[513.3, 1343.8, 1119.6]	[0.1541, 1.0209, -1.2764]

After propagating with the true initial position for the course of a day, the independent code that utilizes the previously discussed approach plots the following output.

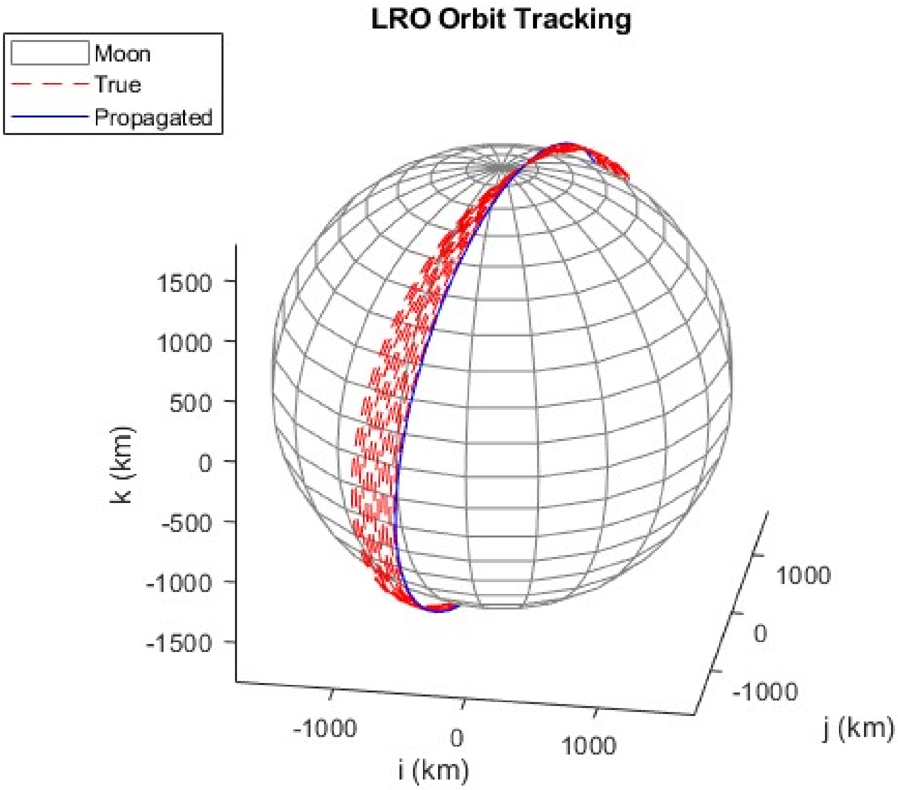

Figure 6: True vs Predicted Orbit Propagation

As can clearly be seen by the above comparison for independent and SPICE propagation, the true orbit propagation experiences precession as the orbit is changing over time which appears to be unaccounted for in the independent propagation method. Thus the final positions and velocities had high error which is shown in the table below.

Table 3: True and Predicted Final Positions and Velocity

TRUE	Date	LRO Position [km]	LRO Velocity [km/s]
True Final	March 16th, 2024 00:00:00.000 UTC	[513.3, 1343.8, 1119.6]	[0.1541, 1.0209, -1.2764]
Predicted Final	March 16th, 2024 00:00:00.000 UTC	[-57.6141, 481.717, -1776.8]	[-0.4780, -1.5029, -0.3908]
Difference	1 day	[570.9, 862.3, 289.62]	[0.6313, 2.5238, -0.8856]

As can be gathered from the above data, the independent code approach failed at providing a good estimate. Many attempts were taken to try and improve this result using knowledge from the course material with little progress. Thus possible source of errors will be discussed as well as potential fixes that could result in future improvements to the final states.

5.1 Comments

The biggest issue and probably the largest reason for the errors in propagation seemed to be coming from the spherical harmonics sha file selection. Three seperate sha lunar files were actually tested in the running of this code however, each one led to vastly different results and run times. It could be likely that there was problem within the text file that caused this as the spherical harmonics code itself was taken directly from previous implementations that worked correctly. As for the LRO mission itself, the major assumption was that the LRO orbiter did not use any major delta v burn during this particular extended mission phase window. As can be seen from the true orbit propagation with the SPICE file ephemeris data, the LRO slowly precesses over the time window which would be due to spherical harmonics of the moon or small thrust burn that are meant to change the orbital plane of the LRO to survey more areas of the moons surface that was not accounted for. Another complication of this code was likely from the enforced ode45 time step as it progressed very slowly when fed the time frame from the SPICE data. This could also be due to a poor interaction between the two libraries. After extensive reading into the SPICE toolkit, I am confident the data is being pulled and read correctly however, the single day propagation take significantly longer than previous course problems.

6 Conclusion

This study set out to compare the ephemeris-based trajectory data from NASA's SPICE toolkit with a custom-built numerical propagation model for the Lunar Reconnaissance Orbiter (LRO), using the modeling techniques and astrodynamics principles taught throughout the semester. While the independent propagation code attempted to use several advanced force models the resulting predicted final state diverged significantly from the true SPICE-derived trajectory. Key limitations identified include potential inaccuracies or incompatibilities in the selected SHA file, numerical integration inefficiencies stemming from the use of MATLAB's `ode45` solver, and possibly unmodeled delta-v maneuvers or perturbations.

References

[1] NAIF-JPL. The spice concept, 2024. Accessed 24 Apr 2025.

[2] NASA. Grail moon lgrs derived gravity science data products v1.0, 2013. Accessed 24 Apr 2025.

[3] NASA. Lunar reconnaissance orbiter (lro), 2022. Accessed 24 Apr 2025.

[4] NASA. Building the lro spacecraft, 2024. Accessed 24 Apr 2025.

[5] Eberhard Gill Oliver Montenbruck. *Satellite Orbits: Models, Methods and Applications*. Berlin, Heidelberg : Springer Berlin Heidelberg : Imprint: Springer, 1st edition, 2000.